

Assignment: Multi-Modal Chest X-Ray Intelligence System (Dual-Mode: Report Generation & QA)

Name: Sama Mohamed Abdelhamid

ID: 202201867

Drive:

https://drive.google.com/drive/folders/1-sxHp9j9PO9TnyuJwR32MnRopx_Fip2n?usp=sharing

Github:

<https://github.com/samaeldomany/Multi-Modal-Chest-X-Ray-Intelligence-System-Dual-Mode-Report-Generation-QA->

1. Architecture Overview

The proposed system implements a two mode medical AI pipeline that can generate a structural Report Generation and Clinical Question Answering. To handle the multi-modal nature of chest X-rays (visual data) and clinical reports (unstructured text), the architecture used a Retrieval Augmented Generation (RAG) framework to prevent the risk of hallucination.

- **Vision-Language Encoder (ColPali):** used for its patch based document understanding to embed both images and text queries into a shared vector space.
- **Vector Database (Qdrant):** the retrieval engine that stores X-ray embeddings with its corresponding clinical reports as metadata.
- **Generative LLM (Gemma):** A 2-billion parameter model strictly prompted to generate reports and answer clinical queries based only on the context retrieved by Qdrant.

2. Technical Challenges and Implemented Fixes

Throughout the development of the multi-modal pipeline, several hardware, data, and constraints were encountered and resolved:

Problem	Architectural Fix
Hardware OOM (Out of Memory) Crashes Loading ColPali and Gemma simultaneously exceeded standard edge GPU limits.	Implemented 4-bit model quantization via BitsAndBytesConfig for Gemma. Utilized device_map={"": 0} and low_cpu_mem_usage=True for direct-to-GPU streaming, bypassing System RAM entirely. Memory was aggressively managed using manual garbage collection.
Messy Dataset Formatting The Kaggle dataset CSV stored image paths as string-encoded arrays ("['path/to/img.jpg]"), breaking basic file matching loops.	Applied Python's ast.literal_eval to safely parse the string arrays, followed by os.path.basename to extract exact filenames. A high-speed dictionary lookup was implemented to instantly link these filenames to the real downloaded .jpg files in milliseconds.
LLM Medical Hallucinations During unconstrained testing, model attempted to infer causes for symptoms	Engineered strict boundary prompts for the RAG pipeline. The LLM was explicitly told to <i>"answer based STRICTLY on the retrieved clinical context."</i> When tested, the system successfully refused to answer questions with information not present in the Qdrant.

3. Dataset Engineering: Synthetic QA Generation

For evaluation of the QA Mode, the subset of the MIMIC-CXR dataset utilized lacked standardized CheXpert label matrices. To solve this, a Zero-Shot Synthetic Data Generation pipeline was used.

Instead of manually hardcoding questions, Gemma was used as a data extractor. It was prompted with strict constraints to read the unstructured clinical reports and generate precise

Question-Answer pairs without using any external medical knowledge.

STRICT RULES:

1. The Answer **MUST** be explicitly stated word-for-word in the text.
2. **DO NOT** guess, diagnose, or infer causes. No external medical knowledge is allowed.

This pipeline successfully yielded my_reliable_vqa_dataset.csv, resulting in a robust, hallucination free ground truth to evaluate the Qdrant retrieval accuracy and generative constraint and human evaluation was used to validate this data.

4. Model Comparison: ColPali vs. PubMedCLIP

To validate the architectural decision of using ColPali for visual text retrieval, a quantitative benchmark was conducted against PubMedCLIP, a domain adapted contrastive model specifically trained on medical literature. The evaluation used a Recall@5 metric across a sample dataset of 200 paired X-rays and reports. The first 200 characters of each clinical report were passed as search queries to see if the models could successfully retrieve the exact corresponding patient X-ray within their top 5 predictions.

- **ColPali** : 6.50%
- **PubMedCLIP** : 3.50%

Chest X-ray images have always been known to be highly homogeneous. The difference between two patients' images may be extremely small that the AI cannot distinguish. Zero-shot retrieval may yield poor performance due to the high density of the information contained in the images. However, in this case, the accuracy of the ColPali model was almost twice that of PubMedCLIP. This shows that despite the fact that a CLIP model can be customized for medical data, it still fails to compare to the ColPali.